Mt. St. HENRY'S HIGH SCHOOL MUKONO

ASPIRATION FOR EXCELLENCE

S.4 PHYSICS SEMINAR HELD ON SUNDAY 28TH, JUNE 2026

THE SEMINAR RESPONSE BOOKLET

COMPILED BY THE PHYSICS DEPARTMENT 2026

WE WISH YOU SUCCESS IN YOUR UNEB NATIONAL EXAMINATIONS MOST ESPECIALLY PHYSICS SUBJECT

INTRODUCTION

This document contains the physics responses to the items for the seminar. The purpose of the seminar is to help you understand effective approaches and techniques for responding to physics examination items, as well as strategies for managing time in both Physics Paper 1 and Paper 2 to achieve excellent performance.

The first section focuses on Physics Paper 1 items arranged according to the different elements of construct, while the final section covers practical physics items. As part of the preparation, we shall begin by guiding you through the breakdown of each element of construct so that you can prepare adequately and confidently for the examination.

We wish you active participation throughout the seminar and every success in your forthcoming UNEB examinations, particularly in Physics.

COMPOSITION OF PHYSICS PAPER 1 (535/1)

Theory

Section A:

This section consists of three (3) compulsory items drawn from three different elements of construct, as shown in the table below.

Each element of construct combines several related topics from the Physics syllabus. Therefore, when preparing for the examination, learners are encouraged to use the table below as a checklist to ensure comprehensive coverage of all the content areas. A good understanding of each element of construct will not only improve content mastery but also enhance confidence when responding to examination items.

As the saying goes, "leave no stone unturned." Thorough preparation across all the elements of construct is key to success in Physics.

S/N	ELEMENT OF CONSTRUCT	TOPICS COVERED	MATHEMATICAL CONCEPT
1	**EARTH, SPACE PHYSICS AND DIGITAL ELECTRONICS.** *Learner understands solar system, galaxies, stars, satellites and communication in everyday life.*	1. The solar system 2. Stars and galaxies 3. Satellite and communication 4. Digital electronics (logic gates)	a) Calculating speed of rotation of orbiting objects like planets, moons and satellites using $speed = \frac{distance}{time}$ b) Calculating the strength of the signals used in satellites using $v = f\lambda$ c) Finding the volume and size of celestial objects

			by assuming them to be **spherical** d) Drawing the truth tables for the possible inputs and outputs of a logic gate system.
2	**ATOMIC MODELS AND NUCLEAR PROCESSES** *The learner understands the structure of atoms, nuclear processes and their application in everyday life.*	1. Atomic models 2. Nuclear processes	a) Finding half-life and time of decay from the decay curve drawn on graph b) Also finding half-life using a formula below $\frac{N_O}{N_T} = (2)^{\frac{t}{T_{\frac{1}{2}}}}$ c) And lastly the chain method, however we encourage that this should end in a graph. d) Calculating the penetration power of the radiations like X-rays by finding frequency/wavelength using $v = f\lambda$.
3	**LIGHT AND WAVES** *Understands how waves are generated, propagated and their applications in everyday life.*	1. Nature of light; reflection of light at plane surfaces 2. Reflection of light at curved surfaces 3. Refraction, dispersion, and colour 4. General wave properties 5. Sound waves	a) Ray diagram construction on graphs for finding the position of the image in both plane and curved mirrors and lens b) Finding power of a lens using $Power = \frac{1}{f(m)}$ c) Finding magnification $M = \frac{v}{u}$ d) For the pin hole camera $\frac{h_i}{h_0} = \frac{v}{u}$ e) The wave equation $v = f\lambda$

Section B

This section is divided into two parts: Part I and Part II.

Each part contains two (2) items. Candidates are required to answer one item from Part I and one item from Part II, making a total of two items answered in this section.

The items in this section often require a deeper understanding of Physics concepts, problem-solving skills, and the ability to apply knowledge to unfamiliar situations. Learners are therefore encouraged to revise all the topics covered under both parts and practise a variety of examination questions to improve their confidence and performance.

PART	ELEMENT OF CONSTRUCT	TOPICS	MATHEMATICAL CONCEPT
I	**MECHANICS AND HEAT** *The learner understands the effect of force and heat on properties of matter.*	1. Measurements in physics especially measurement of length, mass and time 2. States of matter 3. Effects of forces 4. Temperature measurements 5. Heat transfer 6. Expansion of matter 7. Work, energy, and power. 8. Turning effect of forces, centre of gravity and stability. 9. Mechanical properties of materials. 10. Pressure in solids and liquids 11. Linear and non-linear motion 12. Heat quantities and vapours.	a) Finding area formulae of different shapes b) Finding volumes of different regular objects c) Finding density of a substance using ; $\rho = \frac{m}{v}$ d) Finding mass of a substance using principle of moments e) Finding the position of center of gravity using principle of moments f) Conversion between units of temperature g) Finding linear expansivity of a solid using $\alpha = \frac{\Delta l}{l_o \times \Delta\theta}$ h) Finding work done and power using; $W.d = F \times d$ i) Finding power of a machine using; $P = \frac{F \times d}{t}$ or $P = F \times v$ j) Finding efficiency of a machine using; $Efficiency = \frac{power\ output}{power\ input} \times 100$

			k) Finding the moment of a force using; $moment = F \times d$ l) Finding force constant K using; $F = Ke$ m) Finding young's modulus of a material using; $Y = \frac{stress}{strain}$ n) Finding pressure in solids using; $P = \frac{force}{area}$ o) Finding pressure in liquids using; $P = h\rho g$ p) Analyzing motion of a body moving with changing speed using the equations of motion for a body moving in a straight line. q) Analyzing motion of a body moving with constant speed using; $speed = \frac{distance\ covered}{time\ taken}$ r) Analyzing motion of a body using graphs most especially the velocity – time graph s) Finding heat using $heat = mc\Delta\theta$ t) Also using $heat = Pt$
II	**ELECTRICITY AND MAGNETISM** *Appreciates electricity and magnetism in everyday life.*	1. Magnets and magnetic fields 2. Electrostatics 3. Introduction to current electricity 4. Voltage, resistance and ohm's law	a) Finding current, voltage and resistance of a conductor using; $V = IR$ b) Finding power using $P = IV$ c) Find total resistance using **For series network** $R_{eff} = R_1 + R_2 + R_3$ **For parallel network**

		5. Electromagnetic effects 6. Electric energy distribution and consumption	$\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$ d) Finding missing emf, E, and internal resistance using the circuit equation $E = I(R + r)$ e) Calculating the cost of using electricity; $total\ cost = power(kW) \times time(hrs) \times cost\ per\ unit$ f) Finding number of units using; $units = power(kW) \times time(hrs)$

Note: The elements of construct are not given in the order of items in the paper.

COMPOSITION OF PHYSICS PAPERS 2 AND 3 (535/2 & 535/3)

Practical

These papers assess learners in three major assessment areas:

- Mechanics
- Light
- Electricity

The practical paper consists of two (2) items, from which a candidate is required to answer only one item within a period of two (2) hours.

Time Management During the Practical Examination

Effective time management is essential for success in the practical examination. Learners are advised to allocate their time as follows:

- **First 60 minutes:** Carefully read the item, plan for the report, set up the apparatus, perform the experiment, take measurements, and record observations and results.
- **Next 30 minutes:** Process the results, complete calculations, and construct tables and graphs where required.
- **Final 30 minutes:** Analyse the data, interpret the results, draw conclusions, evaluate the procedure, identify sources of error, and suggest appropriate precautions or improvements.

Learners should ensure that all observations are recorded neatly, calculations are clearly shown, and conclusions are supported by the experimental results obtained. Accuracy, organization, and proper interpretation of data are key to scoring highly in the practical examination.

ELEMENT OF CONSTRUCT	TOPIC (s)	COMMON EXPERIMENTS
SCIENTIFIC INVSTIGATION *Appreciate scientific investigation in physics.*	Mechanics	1. Determining unknown mass of a solid 2. Determining unknown mass of a metre rule 3. Determining density of a solid 4. Determining density of a liquid 5. Determining Relative Density (R.D) of a solid/liquid. 6. Determining the force constant of a helical spring
	Light	1. Determining the refractive index of a glass material 2. Determining the critical angle of a glass material 3. Determining the focal length of a concave mirror and convex lens 4. Determining the radius of curvature of a concave mirror and convex lens 5. Determining the power of a concave mirror and convex lens 6. Determining the magnification of a concave mirror and convex lens
	Electricity	1. Determining the resistance of a conductor 2. Determining the internal resistance of a dry cell 3. Determining the e.m.f, E of a dry cell 4. Determining the power dissipated by a conductor or appliance

RESPONSES TO THE ITEMS

Item 1 (a)

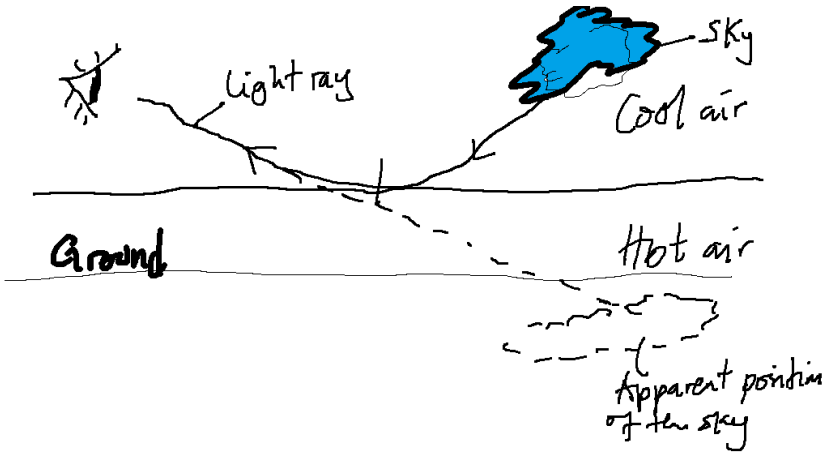
<u>The occurrence along the way to the water reservoir (Mirage)</u>

The air above the short grass appeared like colourless flames of fire due to a mirage.

On a hot day, the heated ground warms up the air immediately above it. The air near the ground becomes less dense than the cooler air above it. Light travelling from the sky through these layers of air is continuously refracted away from the normal line until a point is reached where the angle of incidence becomes greater than the critical angle, causing it to undergo total internal reflection before reaching the boy's eyes.

Since light travels in a straight line, to the boy the light appears as if it is coming from the short grass, which is actually an image of the sky that appears like colourless flames of fire.

Illustration

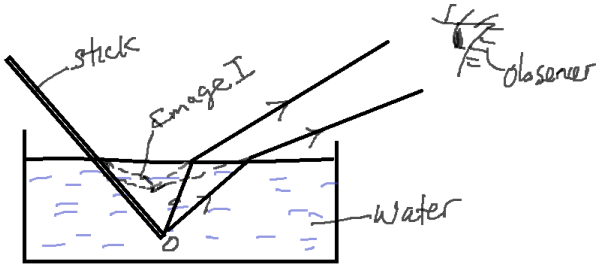


<u>The occurrence at the water reservoir (Refraction)</u>

When part of the stick is immersed in the water, it appears to be broken due to the refraction of light.

Light rays travelling from the denser medium (water) to the less dense medium (air) bend away from the normal line at the boundary (interface). When projected back, these rays appear to originate from a shallower depth, making the part in the water appear bent upwards even though it is actually straight.

Illustration



Item 1. (b)

<u>Cause of the Second Sound</u>

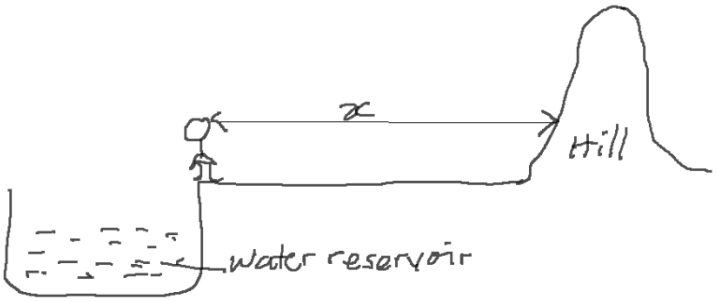
The second sound heard by the boy was an echo.

An echo occurs when sound waves are reflected from a large surface like a hill and return to the listener. When the boy shouted, the sound travelled to the hills, was reflected, and returned to him after 1.5 seconds. This made it seem as though someone was imitating him.

Item 1 (c)

Calculation of Distance to the Hills

Information gathering



Using the echo formula for distance:

$$\boldsymbol{v = \frac{2x}{t}}$$

Given parameters:

Velocity of sound (v) = *320 m/s*

Total time taken (t) = *1.5 s*

Distance to the hills = *x*

Substituting the values into the formula:

$$320 = \frac{2x}{1.5}$$

$$2x = 320 \times 1.5$$

$$2x = 480$$
$$x = \frac{480}{2}$$
$$x = 240\,m$$

Therefore, the distance of the hill from the reservoir is **240 m**

Item 1 (d)

Applications of Refraction and Total Internal Reflection

- Optical fibre communication used in internet and telephone networks.
- Medical endoscopes used by doctors to examine internal body organs.
- Prisms in binoculars and periscopes to reflect light efficiently.

Item 2 (a)

Why sound is clearer and more disturbing at night than during the day.

Propagation During the Day:
During a sunny day, the ground becomes hot and heats the adjacent layer of air. As a result, the air near the ground becomes hotter and less dense, while the air above it remains cool and denser. Since the speed of sound is higher in warmer (less dense) air, sound waves travel faster near the ground. When sound waves travel from a less dense medium to a denser medium, they are **refracted upwards**. Consequently, less sound energy reaches people who are far away at ground level, making the sound appear less intense and less disturbing.

Propagation During the Night:
During the night, the ground loses heat rapidly via radiation, cooling the air layer immediately above it. The air near the ground thus becomes cooler and denser, while the air above remains warmer and less dense. When sound travels from the dense air near the ground toward the less dense air above, it is **refracted downwards** towards the ground. As a result, sound travels to significantly greater horizontal distances along the surface, and more sound energy reaches nearby homes and residents, leading to a higher level of disturbance.

Item 2 (b)

Evaluation of the Owner's Claim

Given Parameters:

- Speed of sound in air (v) = 330 m/s
- Wavelength (λ) = 2.0 cm = 0.02 m

From the wave equation:

$$\boldsymbol{v = f \times \lambda}$$

Rearranging to solve for frequency (f):

$$f = \frac{v}{\lambda}$$
$$f = \frac{330}{0.02}$$
$$f = 16{,}500\ Hz$$

Conclusion & Evaluation:
From the calculation above, the sound produced has a frequency of **16,500 Hz**. While this frequency lies within the human audible range of 20 Hz to 20,000 Hz, the owner's claim that it is completely 'harmless' is **NOT VALID**. This frequency is relatively high and sits near the upper limit of human hearing. Prolonged exposure of nearby residents to high-frequency sounds can cause significant discomfort, sleep disturbances, cognitive fatigue, and potential long-term hearing damage. Therefore, the owner must implement noise control measures to limit the transmission of sound to the surrounding community.

Item 2 (c)

Colour Perception of Costumes under Coloured Illumination

Explanation of the Phenomenon:
The costumes appeared to change colour depending on the wavelength of light illuminating them. In reality, the physical colour of the costumes did not alter; instead, the perceived color depends entirely on the wavelengths of light present in the illumination and which specific wavelengths the surfaces are capable of reflecting or absorbing.

Original Costume Colour	Under Blue Light 🔵	Under Green Light 🟢
Yellow Surface	Appears Dark / Black (Absorbs blue light)	Appears Green (Reflects green light)
Red Surface	Appears Dark / Black (Absorbs blue light)	Appears Dark / Black (Absorbs green light)
Black Surface	Remains Black (Absorbs all colours)	Remains Black (Absorbs all colours)

Item 2 (d)

Strategic Advice to the Entertainment Centre Owner & Local Council

To mitigate environmental noise pollution and ensure residential comfort while maintaining an optimal entertainment atmosphere.

Install Acoustic Barriers: Enclose or surround the entertainment center with professional sound barriers or acoustic absorption panels to dampen escaping audio.

Volume Control Protocols: Programmatically reduce the loudspeaker volume thresholds, particularly during late-night hours.
Environmental Buffers: Plant dense tree lines or construct perimeter tall walls designed to scatter and absorb acoustic energy.
Directional Audio Array: Align and direct loudspeaker arrays strictly toward the audience area and completely away from residential zones.
Curfew Establishment: Partner with the local council to establish and enforce reasonable closing hours for outdoor amplified events.
Regular Monitoring: Implement continuous sound-level monitoring using calibrated decibel meters to guarantee compliance with legal limits.

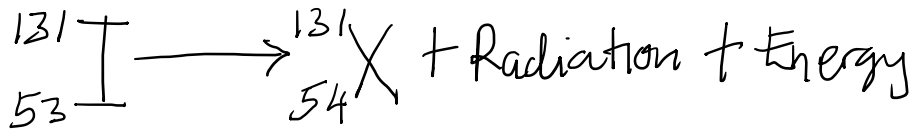
Item 3 (a)

The patient should not be given freedom to go everywhere because Iodine-131 which is highly radioactive was administered. It spontaneously emits ionizing radiation as it decays. Uncontrolled movement of the patient would expose the public and relatives to ionizing radiation, which **damages living cells, breaks chemical bonds in DNA, and causes mutations, leading to cell death or cancerous growths.** The patient is therefore isolated until the activity level of the radioactive substance falls to a safe level close to the background radiation. This minimizes radiation exposure to other people and ensures public safety.

Item 3 (b)

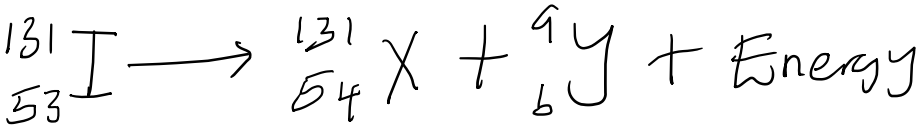
<u>Nature and properties of the radiation Emitted.</u>

The given equation is:



Let the radiation be defined by $^{a}_{b}y$

The equation now becomes



From balancing the mass numbers on both sides of the equation:

$$131 = 131 + a$$

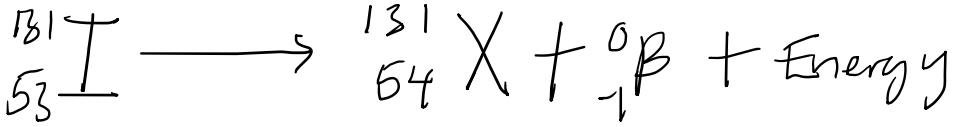
$$a = 0$$

From balancing the atomic numbers on both sides of the equation:

$$53 = 54 + b$$
$$b = -1$$

The Radiation has mass number 0 and atomic number -1 and this is a beta particle. Therefore; the Iodine-131 decays with emission of beta particles (radiation).

The balanced equation becomes:



Properties of beta particles

- They are fast-moving electrons
- They carry a negative charge
- They are moderately ionizing.
- They are more penetrating than alpha particles but less penetrating than gamma rays.
- They can be stopped by a few millimeters of aluminium.
- They are deflected by electric and magnetic fields towards the positive plate.
- They travel at a very high speed close to the speed of light.

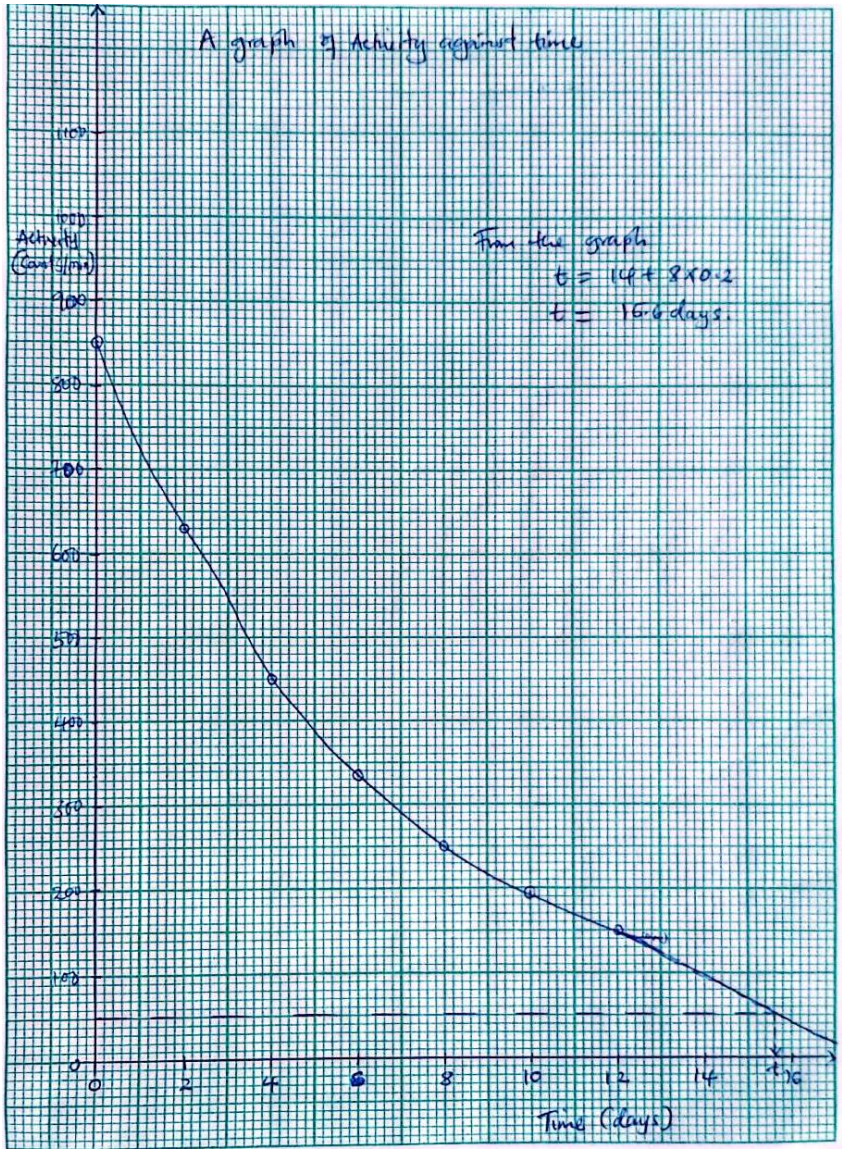
<u>Alternative way of establishing the nature</u>

Looking at the decay equation:



The atomic number increases by 1, while the mass number remains unchanged. This shows Beta decay (β^-).

Item 3 (c)



From the graph; the patient is most likely going to be free after 15.6 days

Item 3 (d)

<u>Three Applications of Radioactive substances similar to Iodine-131.</u>

Medical diagnosis. Radioactive tracers are used to investigate the functioning of organs such as the thyroid gland, kidneys and heart.

Cancer Treatment. Radioactive isotopes are used in radiotherapy to destroy cancerous cells while minimizing damage to surrounding healthy tissues.

Industrial thickness Control. Beta emitting radioactive sources are used to monitor and control the thickness of paper, plastic sheets and metal foils during manufacture.

Other acceptable applications include:

- ✓ Carbon dating of archaeological remains

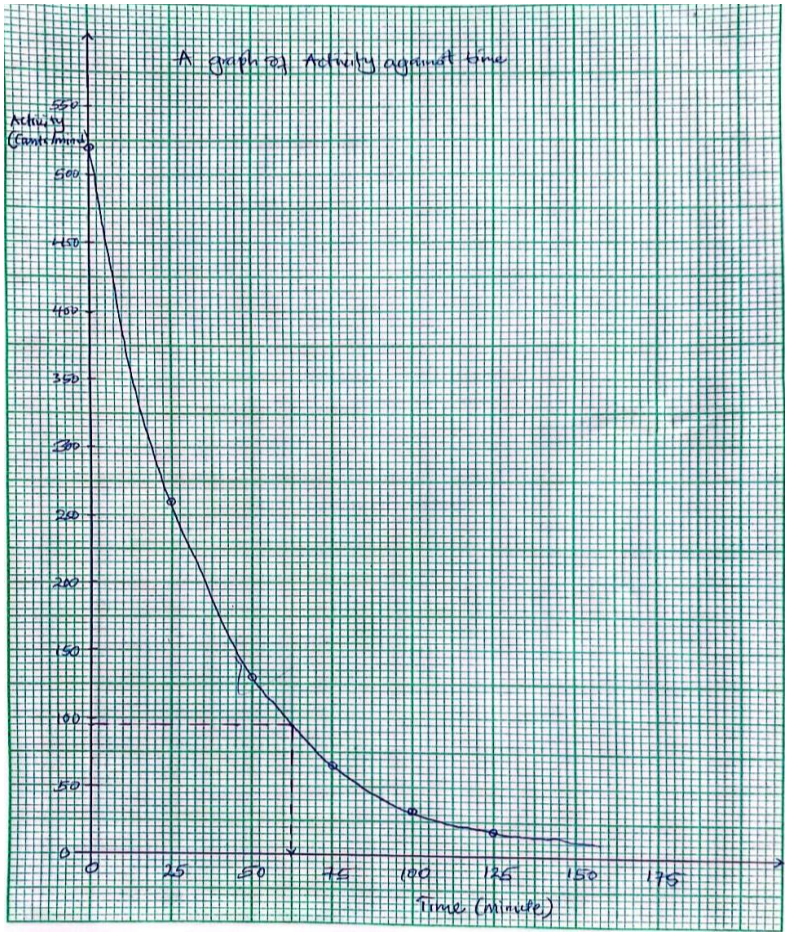
- ✓ Sterilization of medical equipments.
- ✓ Detection of leaks in underground pipelines
- ✓ Food irradiation to preserve food.

Agriculture: utilizing phosphorus-32 tracers to study root absorption rates of fertilizers.

Item 4 (a)

The table below shows how the activity reduces with time

Activity (counts per min)	520	260	130	65	32.5	16.25
Time (minutes)	0	25	50	75	100	125



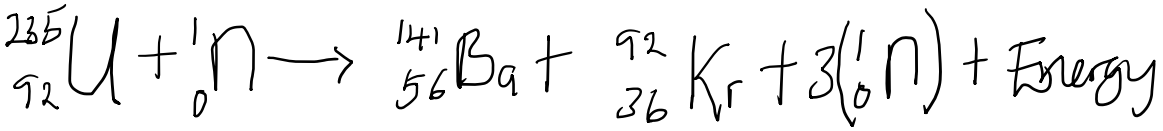
From the graph; they can be allowed to enter the affected section after 62.5 minutes.

Item 4 b (i)

<u>Nuclear process responsible for power generation.</u>

The nuclear process responsible for energy production in nuclear power station is **nuclear fission**.

In nuclear fission, a Uranium-235 bombarded with a slow neutron and splits into two lighter nuclei while releasing large amounts of energy and two or more neutrons. The equation of the reaction is given below.



The emitted neutron can cause further fission reactions, producing a Controlled chain reaction resulting into a lot of energy.

Item 4 b (i)

<u>Conversion of Nuclear Energy into Electrical energy.</u>

The energy produced from nuclear fission reaction is heat energy which is used to turn water into high-pressure steam.

This steam rotates a turbine, the turbine drives a generator and the generator converts mechanical energy into electrical energy through electromagnetic induction.

Then finally the electricity is transmitted through power lines to consumers.

Note: Energy Transformations.

Nuclear Energy → Heat Energy → Mechanical energy → Electrical energy

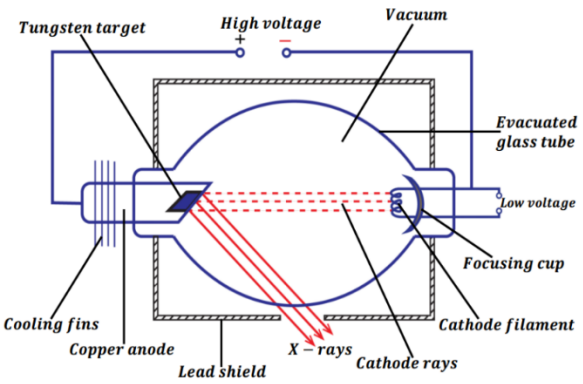
Item 4 (c)

Comparison of Alpha, Beta and Gamma in terms of.

Penetrating power:
Alpha is very low (stopped by paper);
Beta is moderate (stopped by thin aluminium sheet);
Gamma is high (requires thick lead / concrete shields).
Ionizing Ability:
Alpha is extremely high (due to +2 charge and heavy mass);
Beta moderate;
Gamma is very weak (neutral electromagnetic radiation)
Fields:
Alpha deflects slightly toward the negative plate;
Beta deflects strongly towards the positive plate;
Gamma passes straight without deflection.

Item 4 (d)

X-ray production



Inside an x-ray tube, a cathode filament is heated by a low-voltage supply to emit electrons via **thermionic emission**

A high voltage (10-100kV) accelerates these electrons towards a target metal (tungsten) anode. When the fast-moving electrons strike the target, they are suddenly decelerated. 99% of their K.E is converted into heat and the remaining 1% is converted into x-rays.

Thus, X-rays are produced when high-speed electrons strike a metal target and undergo a rapid deceleration.

Item 4 (e)

Denying learners access prevents acute radiation exposure, which causes Cellular damage and increase Cancer risks because radioactive materials had been released in the affected section

Exposure to ionizing radiation can:

- Damage body cells
- Cause tissue damage
- lead to genetic mutation
- Increases risks of Cancer.

Allowing them would expose them to unnecessary the above hazards

Precautions

- Spending the shortest possible time near radioactive sources
- Staying as far as possible from radiation sources
- Using lead, concrete and other shielding materials to absorb radiation.
- Following strict safety procedures
- Using remote handling equipment where necessary
- Wearing protective clothing.

Item 5 (a)

Appearance of the Moon.

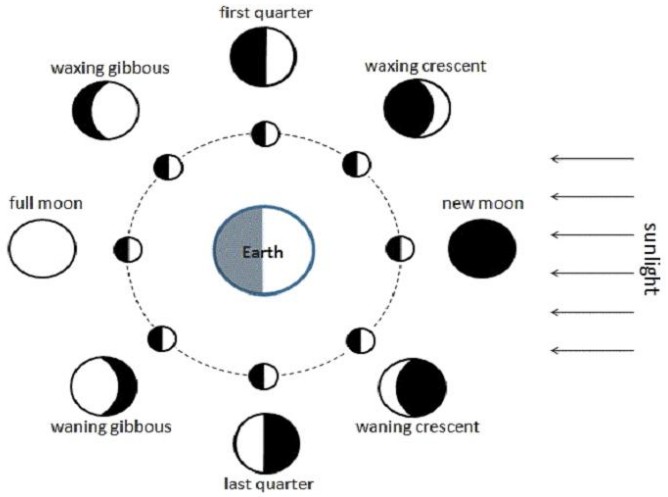
The moon does not produce its own light. It appears bright because it reflects light from the sun.

The bright and dark sections of the moon are due to the fact that, at any time, only a certain portion of the moon is illuminated by the sun.
As the moon revolves around the earth, different portions of the illuminated part become visible from Earth.
As a result, the moon appears to have bright and dark sections.

Phases of the Moon

The phases of the moon occur because of the changing positions of the Moon as observed from the earth.
The main phases are; New moon, waxing Crescent, First quarter, waxing Gibbous, Full moon, Waning Gibbous, Last quarter, and waning Crescent.
One complete Cycle of phases of the moon takes about 29.5 days.

Illustration of the phase of the moon



Item 5 (b)

<u>Cause of the difference between in Lunar and Gregorian</u>

Lunar Calendar based on the motion of the Moon around the Earth.
One lunar month lastly for ~ 29.5 days
But a year has 12 months.
Therefore; One lunar year contains ≈ 12 × 29.5 ≈ 354 days
Gregorian (Solar) calendar is based on the revolution of the Earth around the Sun.
One complete solar year lasts for about 365.25 days
Thus: 1 solar year = 365.25 days.
Difference = 365.25 – 354 ≈ 11.25 days
Thus, the Lunar Calendar is about 11 days shorter than the Gregorian calendar each year.
This is because the lunar Calendar depend on the moon's revolution around the earth, and the Gregorian calendar depends on the earth's revolution around the Sun.

Item 5 (c).

<u>Star appearance</u>

(i) Different colours: stars appear with different colours because they have different surface temperature

• Red stars are relatively Cooler
• Yellow stars because intermediate temperatures
• Whites are hotter
• Blue Stars are the hottest.

(ii) Different Brightness: stars appear with different brightness because they have different sizes, they emit different amounts of energy and they are at different distances from Earth. Large or nearby stars appear brighter than small or distant stars.

(iii) Twinkling of stars. Stars appear to twinkle because their light passes through layers of the Earth's atmosphere having different densities

As the light passes through these layers, it is continuously refracted - The changing refraction causes variations in the apparent position and brightness of the stars, making them appear to twinkle.

Item 5 (d)

Type of the logic gate used and truth table.

The alarm sounds only when Smoke is present and Temperature is high.

This condition requires both inputs (condition) to be satisfied at the same time, therefore, the logic gate used is an AND gate.

Operation of the AND gate.

The AND gate produces an output of 1 only when all inputs are 1.

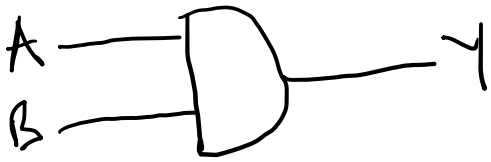
Let:

Input A = Smoke detected

Input B = High temperature detected

Output Y = Alarm.

No_inputs = $2^n = 2^2 = 4$



Truth table

Input A	Input B	Output Y
0	0	0
0	1	0
1	0	0
1	1	1

From the table, it is clear that the alarm sounds only when Smoke is present and Temperature is high. All the other inputs the alarm is off.

Item 6 (a)

The learners were enjoying a bright, warm afternoon at school while the Country shown on the tablet was experiencing Snowstorms, freezing temperatures due to the **Earth's tilt and its revolution around the sun.**

The Earth is tilted at an angle of 23.5° to its orbit. As it revolves around the sun, different regions receive different amounts of solar energy at different times of the year.

The country shown in the news broadcast was likely experiencing winter, where the sun's rays strike the Earth's surface at a smaller angle. As a result, less solar energy is received, temperatures became very low, and snowstorms may occur.

On the other hand, the learners' location was receiving more direct sunlight. This caused higher temperatures, resulting in the warm conditions they were experiencing during lunch break.

Therefore, the country was not cold because it was farther from the sun as some learners suggested, but because it was receiving less concentrated solar energy due to the Earth's tilt and revolution.

Item 6 (b)

The learners were watching the broadcast during a bright afternoon while the people in the distant country were experiencing darkness due to the rotation of the Earth on its axis.

The Earth rotates from west to east once every 24 hours. At any moment, one half of the earth faces the sun and experiences day, while the other half faces away from the sun and experiences night.

At the time of the broadcast, the school was on the side of the Earth facing the sun, which is why the learners where outside enjoying daylight. However, the country shown on the tablet was on the side facing away from the sun, resulting in nighttime condition.

This shows that the sun does not shine only on Africa as some learners believed, instead, the different parts of the earth face the sun at different times because of the Earth's rotation.

Item 6 (c)

Pictures and sound from such a distant country reach the Learner's tablet through the application of **electromagnetic waves** and **communication Satellites.**

Audio and video at the source are converted into electrical signals by microphones and cameras.

These signals are modulated onto high-frequency microwaves or radio waves and sent to geostationary communication Satellites by a ground uplink station.

Satellites receive, amplify, and retransmit the signals back down-to-Earth stations.

The signals are distributed via local transmitters or fibre-optic cables to the tablet's receiver, which them back into sound and Images.

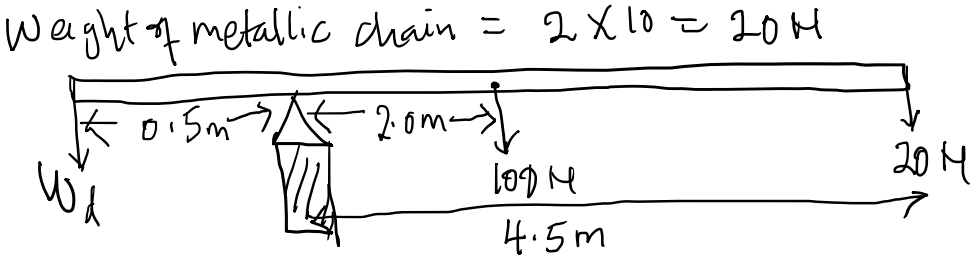
Item 6 (d)

<u>Applications of micro waves / Radio waves.</u>

- ✓ Global positioning system (GPS) Satellite navigation.
- ✓ Radar systems for aviation tracking and weather forecasting
- ✓ Rapid heating of food in domestic microwave ovens.

Item 7 (a).

Determining whether money given to askari will be enough.



Let the weight of rock dust be W_d

By taking moments at the pivot;

Sum of Clockwise moments = 100 × 2.0 + 20 × 4.5

= 200 + 90

= 290 Nm

Anticlockwise moment = W_d × 0.5 = 0.5 W_d Nm

Applying the principle of moments at the pivot

Sum of clockwise moment = sum of anticlockwise moments.

290 = 0.5 W_d

W_d = 580 N.

Finding the mass of rock dust.

$$From\ W_d = m_d\ g$$
$$W_d = 580\ N, g = 10\ m/s^2$$
$$580 = 10 \times m_d$$
$$m_d = 58\ kg$$

Finding the total money required.

$$1\ kg = shs.500$$

$$58\,kg = (58 \times 500)$$
$$= 29000\,/=$$

From the calculation above, the total money required for the enough rock dust is 29000 which is higher than the money that was given to askari by 4000/=. Therefore, the money given to him will not be enough they must add him 4000/= to achieve the target.

Item 7 (b)

Determining the specific heat capacity of sugar.

Heat lost by tea = Heat gained by sugar

$$m_t \times c_t \times (\theta_3 - \theta_2) = m_s \times c_s \times (\theta_2 - \theta_1)$$

Let the initial temperature of sugar be the room temperature of 20°C

Where:
m_t = mass of tea
c_t= Specific heat capacity of tea
m_s = mass of Sugar
c_s= specific heat capacity of sugar

Finding mass of tea.

From

$$m_t = \rho \times V$$
$$V = 0.4\,l = (0.4 / 1000)\,m^3 = 0.0004\,m^3$$
$$\rho = 1200\,kg\,m^{-3}$$
$$\Rightarrow m_t = 0.0004 \times 1200 = 0.48\,kg$$

Given values:

$$m_t = 0.48\,kg, \quad c_t = 4000\,J/kgK, \quad m_s = 100g = 0.1\,kg$$
$$\theta_1 = 20°C, \quad \theta_3 = 80°C, \quad \theta_2 = 60°C$$

Substituting the values;

$$0.48 \times 4000 \times (80 - 60) = 0.1 \times c_s \times (60 - 20)$$
$$0.48 \times 4000 \times 20 = 0.1 \times c_s \times 40$$
$$38400 = 4 \times c_s$$
$$c_s = \frac{38400}{4}$$
$$c_s = 9600\,J\,Kg^{-1}\,K^{-1}.$$

Therefore, the specific heat capacity of sugar is $9600\,J\,Kg^{-1}\,K^{-1}$.

Assumption made

• Sugar was initially at room temperature (20°C)
• No heat was lost to the surrounding.
• No heat was absorbed by the cup and spoon.
• Heat lost by tea equals heat gained by sugar.

Item 7 (c)

Determining the rate of cooling in 30 minutes

$$pt = m_t \times c_t \times (\theta_2 - \theta_1)$$

$$p \times 30 \times 60 = 0.48 \times 4000(60 - 45)$$

$$p = \frac{28800}{1800}$$

$$p = 16W$$

Now getting the temperature after 40 minutes

$$pt = m_t \times c_t \times (\theta_2 - \theta_1)$$

$$16 \times 40 \times 60 = 0.48 \times 4000(60 - \theta)$$

$$(60 - \theta) = \frac{38400}{1920}$$

$$\theta = 40\ ℃$$

Therefore, the temperature of tea after 40 minutes is 40 ℃

Item 7 (d)

The tea cooled rapidly after preparation because;

The cup was black: Black Surfaces are very good emitters of thermal energy (radiation) and therefore lose heat rapidly by radiation
Large temperature difference. Initially the tea was at 80°C, which is much hotter than the surrounding air at room temperature. A large temperature difference increases the rate of heat loss.
Metallic Cup: the metallic cup is a good conductor of heat. Heat was conducted through the cup walls to the surrounding.
Convection: Air surrounding the Cup was heated and rose away. Cool air continuously replaced it, increasing heat loss by convection.
Evaporation: Some water molecules escaped from the tea surface as steam. Evaporation removes latent heat and causes cooling.
Addition of sugar and stirring: The sugar and metallic spoon absorbed heat from the tea, causing an immediate drop in temperature from 80°C to 60°C

Item 7 (e)

Applications of Blackened surfaces in Everyday life.

Since black surfaces are excellent absorbers and emitters of thermal radiation, they are applied in:

- ✓ Solar water heaters to absorb heat from the sun and boil the water.
- ✓ Solar cookers
- ✓ Solar panels
- ✓ Blackened radiators in Cars as cooling systems
- ✓ Interior surfaces of thermos flask.
- ✓ Heat collectors in industrial solar energy systems

Item 8(a)

Calculating the gravitational potential energy gained.

$$\boldsymbol{G.P.E = m \times g \times h}$$

Let; $\boldsymbol{m} = Total\ mass\ of\ the\ truck\ and\ half\ a\ tank\ of\ water.$

$$\boldsymbol{m = m_t + m_w}$$

Finding mass of water; $\boldsymbol{m_w}$

$$m_w = \rho \times V$$

$$\rho = 1000\ kg\ m^{-3}, V = \frac{1500}{1000} = 1.5\ m^3$$

$$m_w = 1000 \times 1.5 = 1500\ Kg$$

$$m_w = 1500\ Kg\ and\ m_t = 5000\ Kg$$

$$m = 1500 + 5000 = 6500\ kg$$

Substituting the following values;

$m = 6500\ Kg,\ g = 10\ ms^{-2}, h = 40\ m,$

$$\Rightarrow G.P.E = 6500 \times 10 \times 40 = 2{,}600{,}000\ J\ or\ 2.6 \times 10^6\ J\ or\ 2.6MJ$$

From the calculations above, the truck gained a gravitational potential energy of 2.6×10^6 J.

Item 8(b)

First stage (Acceleration)

u = 0, V = 90kmh^{-1} = $\frac{(90 \times 1000)}{3600}$ = 25ms^{-1}, t_1 = 5minutes = 5 × 60 = 300s

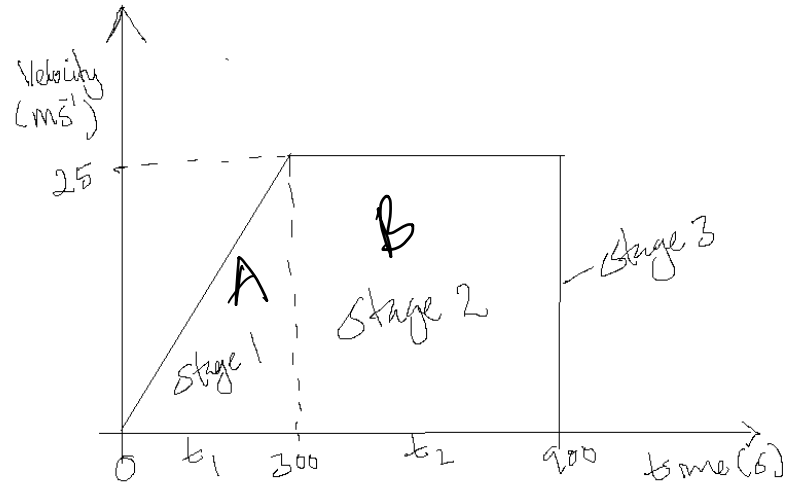
Second stage (Constant speed)

u = 25ms^{-1}, v = 25ms^{-1}, t_2 = 10 minutes = 10 × 60 = 600seconds

Third stage (deceleration)

u = 25ms^{-1}, v = 0, t_3 = 0

Velocity-time graph.



$$\boldsymbol{Total\ distance\ Covered\ =\ Total\ area\ under\ the\ graph.}$$

$$= Area\ of\ A\ +\ Area\ of\ B.$$

$$= ½ \times 300 \times 25 + 600 \times 25 = 3750 + 15000 = 18750\,m$$

Total time = 300 + 600 = 900 seconds

$$Average\ Speed\ (velocity) = \frac{Total\ distance}{Total\ time} = \frac{18750}{900} = 20.83\ ms^{-1}$$

Therefore, the average velocity is 20.83 ms^{-1}

Item 8(c)

Cause of the Smoke and the vigorous shaking

The smoke observed when the brakes were applied was due to friction between the tyres and the road surface. When the driver suddenly stopped, the tyres tended to skid on the road, friction converted kinetic energy into heat, tyres become very hot and dust particles and tyre rubber from the tyre were burned, producing smoke.

The vigorous shaking occurred because the truck was carrying water - when the truck stopped suddenly, the water continued moving forward due to inertia. water then moved backward and forward repeatedly inside the tank. This repeated motion caused the truck to oscillate about its equilibrium position hence the shaking which almost caused an accident.

Therefore, he should avoid sudden braking of the truck when the tank is partially filled as the shifting weight can easily cause a rollover accident.

Item 8(d) (i)

Mass of water, $\boldsymbol{m_w = 1500\ Kg}$

Heat absorbed by water

$$H_{gained} = m_w c_w\,\Delta\theta = 1500 \times 4200 \times (70 - 20) = 315{,}000{,}000\,J$$

Item 8(d) (ii)

Let the total heat produced by the saw mill be H

$$\boldsymbol{\eta = \left(\frac{heat\ absorbed\ by\ water}{total\ heat\ produced\ by\ the\ saw\ mill}\right) \times 100\%}$$

Given information

η = 80%, $heat\ absorbed\ by\ water$ = 315,000,000,

$total\ heat\ produced\ by\ the\ saw\ mill$ = H.

Substituting in the formular; we get

$$\Rightarrow 80\% = \left(\frac{315{,}000{,}000}{H}\right) \times 100\%$$

$$H = \frac{31{,}500{,}000{,}000}{80}$$

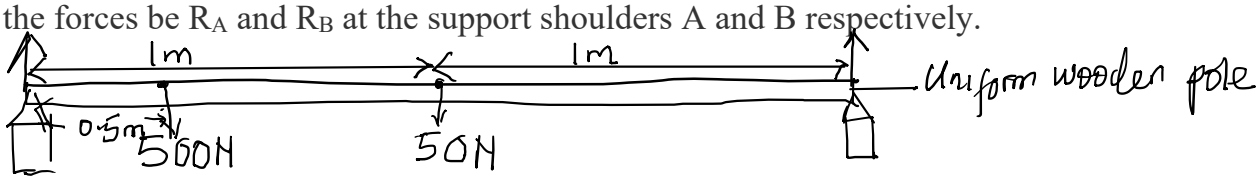
$$H = 393{,}750{,}000\ J$$

Therefore, the total heat produced by the saw mill is $393{,}750{,}000\ J$

Item 9(a)

Determining the forces that caused pain at their shoulders.

Let the forces be R_A and R_B at the support shoulders A and B respectively.



Weight of meat, w = mg = 50 × 10 = 500 N

For vertical equilibrium;

$$Sum\ of\ upward\ forces = Sum\ of\ downward\ forces$$

$$R_A + R_B = 500 + 50$$

$$R_A + R_B = 550 \quad ----(i)$$

Taking moments about support A.

$$Sum\ of\ clockwise\ moments = 500 \times 0.5 + 50 \times 1 = 250 + 50 = 300\ Nm$$

$$Sum\ of\ anti\ clockwise\ moments = R_B \times 2 = 2R_B$$

Applying the principle of moments,

$$Sum\ of\ clockwise\ moment = Sum\ of\ anti\ clockwise\ moment$$

$$300 = 2R_B$$

$$\frac{2R_B}{2} = \frac{300}{2}$$

$$R_B = 150\ N$$

From equation (i)

R_A + **150 = 550**

R_A **= 550 - 150**

$R_A = 400\ N$

From the calculation above, Climber A supported 400H, while the other supported 150 N. These large upward reaction forces caused the tension and pain felt at their shoulder.

Item 9(b)

Finding the time for the meat to be ready.

Assuming no heat lost to the surrounding

$$Heat\ supplied\ by\ the\ heater\ =\ Heat\ gained\ by\ the\ meat\ surrounding\ by\ the$$

$$\boldsymbol{Pt\ =\ m_m\, c_m\, \Delta\theta}$$

$$P\ =\ 750W, \Delta\theta\ =\ 120°c, \qquad m_m\ =\ 50Kg, c_m\ =\ 16500\,J\,Kg^{-1}\,K^{-1}, t\ =??$$

$$750\ \times\ t\ =\ 50\ \times\ 16500\ \times\ 120$$

$$750\,t\ =\ 99{,}000{,}000$$

$$\frac{750\,t}{750} = \frac{99{,}000{,}000}{750}$$

$$t\ =\ 132000\ s.$$

Converting to minutes.

$$1\ minute\ =\ 60\ seconds$$

$$1\ Second\ =\ \frac{1}{60} minutes$$

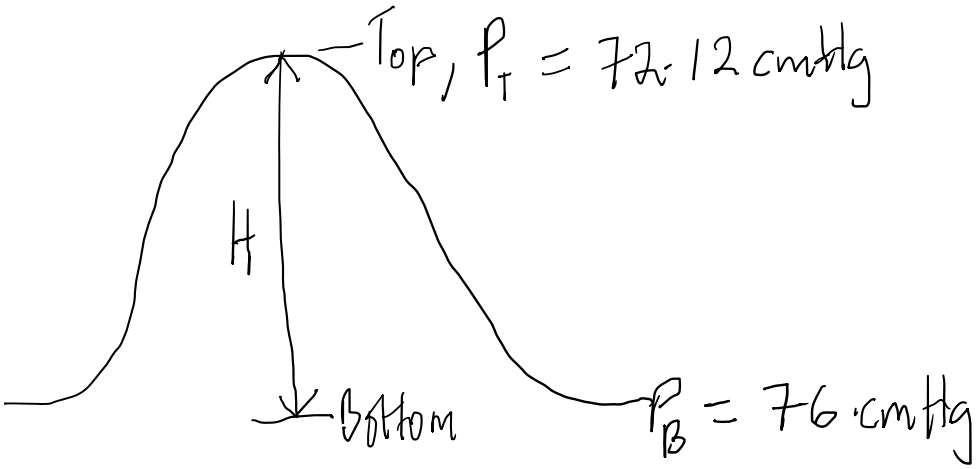
$$132\ 000\ seconds\ =\ \left(\frac{132\ 000}{60}\right) minutes =\ 2200\ minutes$$

From calculation above, the time required for the meat to be ready is 2200 minutes and yet they must deliver the meat in 30 minutes, therefore the climbers will not deliver the meat in time if they decide to eat first.

They should first take the delivery and then come back to eat.

Item 9(c)

Finding the Height of the mountain



Converting Barometric height to pascals

$$P_B\ =\ h_B\,\rho\,g\ =\ \left(\frac{76}{100}\right)\ \times\ 13600\ \times\ 10\ =\ 103360\ Pa$$

$$T_T\ =\ h_T\,\rho\,g\ =\ \left(\frac{72.12}{100}\right)\ \times\ 13600\ \times\ 10\ =\ 98083.2\ Pa$$

$$Pressure\ difference =\ P_B - T_T\ =\ 103360\ -\ 98083.2 =\ 5276.8\ Pa$$

$$Pressure\ of\ air\ Column\ =\ h_{air}\ \times\ \rho_{air}\ \times\ g$$

$$But\ h_{air} = H, \rho_{air} =\ 1.2\ kg\ m^{-3}, g\ =\ 10\ ms^{-2}$$

$$Pressure\ of\ air\ Column\ =\ H\ \times\ 1.2\ \times\ 10 =\ 12H\ Pa$$

But; $\boldsymbol{Pressure\ of\ air\ column\ =\ Pressure\ difference}$

$$12H\ =\ 5276.8$$

$$H\ = \frac{5276.8}{12}$$

$$H\ =\ 439.73\,m\ \approx\ 440m.$$

The estimated height of the hill is approximately 440 m.

Item 9(d)

Climbing to high altitudes can result into nose bleeding because of **pressure difference.**

The atmosphere reduces with altitude which makes the pressure outside the climber's body to drop, but his internal blood pressure remains high.

This pressure difference causes the delicate blood vessels (capillaries) inside his nose to swell and rupture.

Alternatively;

On top of a mountain, atmospheric pressure is lower than that at the bottom. Due to the body's metabolism, the blood pressure may exceed the low atmospheric pressure at the top of the mountain. Since the blood capillaries are weaker, they may break due to the high pressure of the blood thus causing nose bleeding.

Item 9(e)

The wooden block did not break because it has a high tensile and compressive strength and a thick cross-sectional area

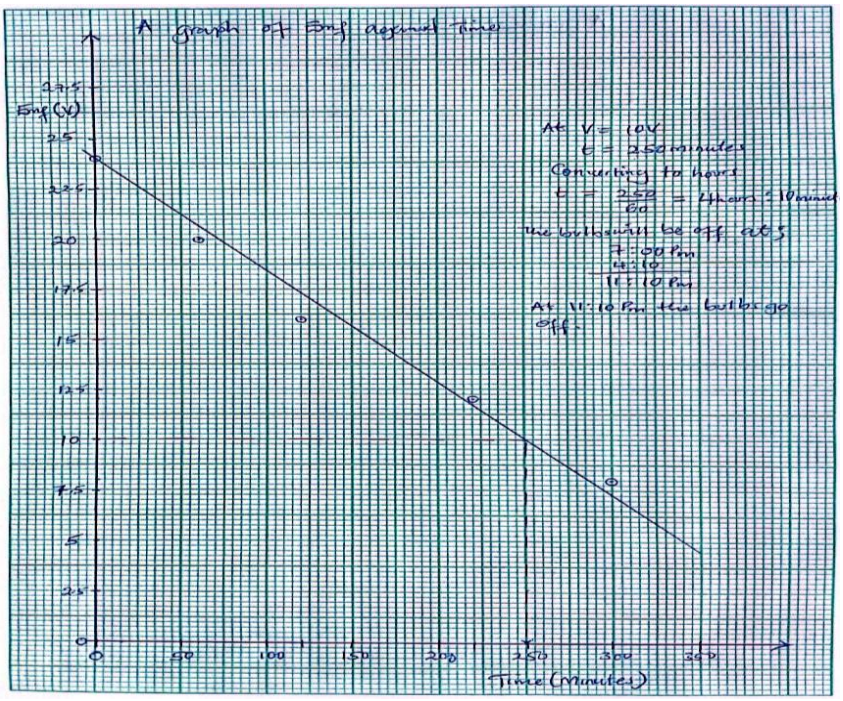
The maximum bending moment exerted by the 500M load did not exceed the structural limits of the thick wooden pole.

In other words, It is strong enough to withstand the breakage force.

Alternatively;

The wooden pole did not break because it was in equilibrium. That is; The sum of clockwise moments about a given fixed point is equal to the sum of ant-clockwise moments about the same fixed point. The sum of forces in one direction is equal to the sum of force in the other direction.

Item 10(a)



From the graph, the bulbs go off at 11: 10 pm.

Item 10(b)

For the system on the parent's house to cook well, Kato must install a solar battery to store energy during day so that it can power appliances at night.

He can also consider connecting all the appliances in parallel circuit since this kind of circuit allows other appliances to continue working in case one gets a problem.

Item 10(c)

Determination of electrical power and current of the Kettle.

For current,

$$I = \frac{V}{R}$$

$$v = 240 \; and \; R = 12 \, \Omega.$$

$$\Rightarrow I = \frac{240}{12}$$

$$I = 20A$$

For Power,

$$P = IV.$$

$$P = 20 \times 240$$

$$P = 4800W$$

The power of the kettle and the current it uses are 4800W and 2A respectivcely.

Item 10(d)

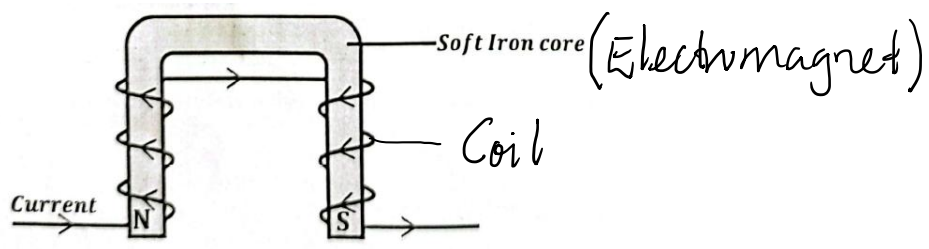
Electricity is used to operate:

- water pumps
- Electric motors
- Lifts
- Factory machines
- Fans

Item 11(a)

The device uncle should use is an **electromagnet (lifting magnet)**

Structure of a lifting magnet



When the switch is closed, current flows through the coil, the soft iron core becomes magnetised and attracts magnetic materials such as iron and steel.

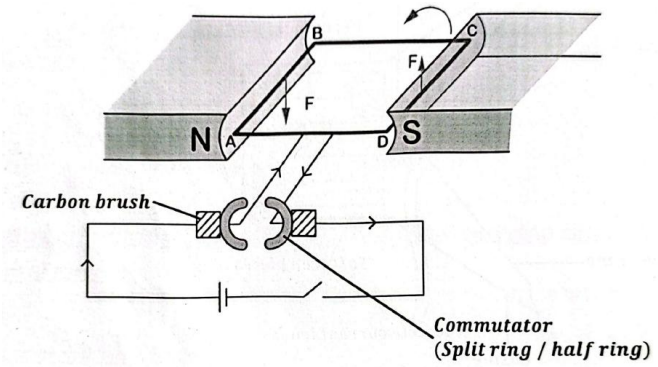
It can pick rusty metals which are magnetic materials in nature by inducing magnetism in them.

It cannot remove glass because glass is a non-magnetic material. Since it cannot be magnetised, it is not attracted by a magnetic field.

Therefore; the electromagnet is suitable for removing the rusty metal pieces but not the glass because glass is non-magnetic.

Item 11(b)

Structure of a motor



a) When the switch is closed, current flows through a rectangular coil, ABCD of wire inside the motor
b) When current flows through the coil, it produces its own magnetic field.
c) The magnetic field of the coil interacts with the magnetic field of the permanent magnets, causing the coil to experience a force (this called motor effect)
d) Side CD experiences an upward force and side AB experiences a downward force according to Fleming's left-hand rule.
e) The two forces form a couple which causes the coil to rotate in the anticlockwise direction.
f) When the coil reaches the vertical position, the carbon brushes lose contact with the commutators and current is cut off.
g) However, the coil continues to rotate past the vertical position due to momentum gained.
h) The two commutators interchange contacts with the carbon brushes. This reverses the direction of current in the coil and the forces experienced by the sides of the coil.
i) The rotating coil is connected to the milling machine and hence enabling it to spin in one direction.in one direction.

Item 11(c)

The noise and heat are due to the existence of frictional resistance.

The worn-out bearings cause heavy friction, which generates loud noise and forms mechanical energy into unwanted heat.

This friction can be reduced by applying grease or replacing the bearings.

Determination of the efficiency (η)

$$\boldsymbol{Efficiency\ (\eta) = \left(\frac{Useful\ work\ output}{work\ input}\right) \times 100\%}$$

$$Useful\ work\ output\ = 2.7 \times 10^6\ J\ = 2700000\ J$$

$$work\ input\ (Electrical\ energy\ supplied\ energy)\ = IVt$$

$$I\ = 30A, \qquad V\ = 240, \qquad t\ = 10\ minutes\ = 600\ seconds$$

$$\Rightarrow work\ input\ =\ 30\ \times\ 240\ \times\ 600 =\ 4{,}320{,}000\,J.$$

$$\therefore\ \eta\ =\ \left(\frac{2700000}{4320000}\right)\times\ 100 =\ 62.5\%$$

The efficiency is 62.5 which is very small indicating a very high loss of energy as wasted energy to overcome friction in the machine.

Item 11(d)

Uses of motors

They are used in;

- Electric fans
- water pumps
- Refrigerators
- Electric Vehicles
- Elevators and lifts
- Industrial machines.
- Blenders and mixers.
- Washing machines

It operates by converting electrical energy into mechanical energy.

Item 12(a)

Current needed in the radio, $\boldsymbol{I_r}$

$$\boldsymbol{I_r\ =\frac{P_r}{V_r}}$$

$$P\ =\ 90W,\qquad V\ =\ 120V$$

$$\Rightarrow\ I_r\ =\ \frac{90}{120}\ =\ 0.75A$$

Current need in the kettle, I_k

$$\boldsymbol{I_k\ =\frac{P_k}{V_k}}$$

$$P_k\ =\ 2400\,V,\qquad V_k\ =\ 120\,V.$$

$$\Rightarrow\ I_k\ =\ \frac{2400}{120} =\ 20A$$

Total current I drawn by the two appliances

$$I\ =\ I_r\ +\ I_k$$

$$=\ 0.75\ +\ 20$$

$$=\ 20.75A$$

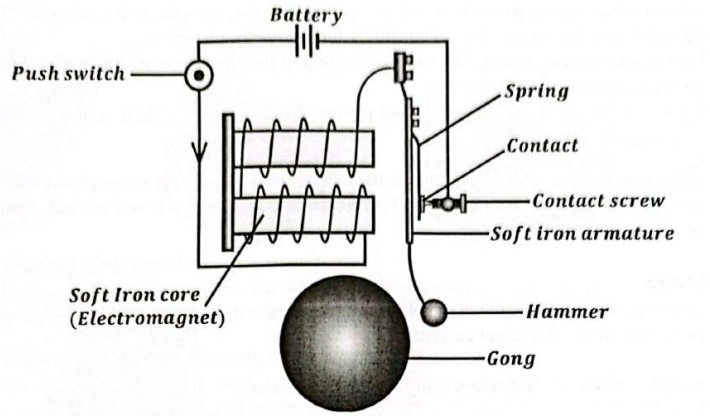
From Calculation above, the current drawn by the two appliances is 20.75A which is greater than 10A of the fuse, Therefore the fuse will blow if it used to protect them.
They should get another one of high current rating.

Item 12(b)

The six lamps should be connected in parallel. This is because; each bulb receives a full supply of voltage, each bulb operates at its rated brightness, if one bulb fails, the others continue working and each lamp can be controlled independently.

Item 12(c)

Structure of an electric bell



When the switch is pressed, current flows through the coil, Coil becomes an electromagnet and attracts the armature. Hammer strikes the gong. Contact breaks and current stops. Electromagnet loses magnetism and the armature returns to its original position. Process repeats rapidly, producing ringing sound.

Item 12(d)

Cost of using electricity

Cost for operating a radio 8 hours in a day

$$Cost \ = \ Power \ (Kwh) \ x \ time \ (h) \ x \ cost \ per \ unit$$

$$P \ = \ 90W \ = \ \frac{90}{1000} \ KW$$

$$t \ = \ 8 \ hours$$

$$cost \ per \ unit \ = \ 750 \ /=$$

$$Cost = \left(\frac{\mathbf{90}}{\mathbf{1000}}\right) \times \mathbf{8} \times \mathbf{750} = \mathbf{540} \ /=$$

Cost for operating six lamps for 5 hours in a day

P = 100 W (1 bulb)

Total power of 6 bulbs; $P \ = \ 100 \ \times \ 6 \ = \ 600W \ = \ \frac{600}{1000} KW$

T = 5 hours

$$Cost \ = \left(\frac{600}{1000}\right) \times \ 5 \ \times \ 750$$

$$cost \ = \ 2250 \ /=$$

$$Total \ cost \ for \ operating \ both = \ 540/= \ + 2250/= \ \mathbf{2790}/=$$

Therefore; the daily cost of operating the radio and the six lamps is 2790.

RESPONSES TO THE PRACTICAL ITEMS

ITEM 1

AIM

An experiment to determine if the force constant of the spring is in the range of $200Nm^{-1}\ to\ 600Nm^{-1}$, so as to replace the damaged one in the punching machine.

VARIABLES

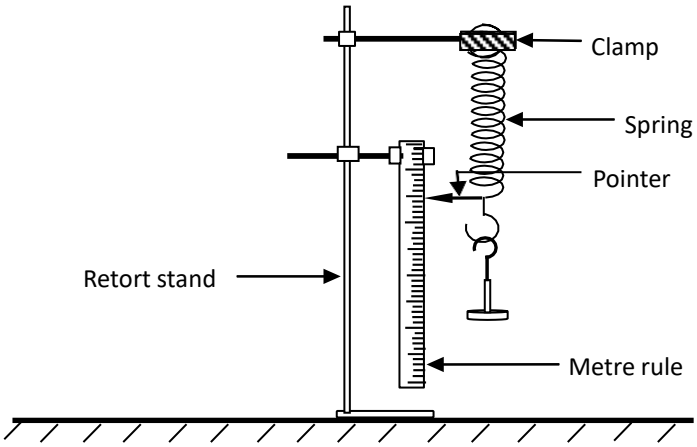
Variable	Example (s)
Independent variable	Mass of the standard/slotted mass, Weight of the standard/slotted mass.
Dependent variable	Extension / change in length, e/ New position of pointer, P_1/ New length of the spring, l
Controlled variable	Original length of the spring/ Initial position of the pointer,P_0/ Mass of the spring / Number of turns on the spring

HYPOTHESIS

The force constant of the spring is in the range of $200Nm^{-1}\ to\ 600Nm^{-1}$

PROCEDURE

(i) Experimental setup/diagram;



(ii) Steps in performing the experiment;

a) The apparatus were arranged as shown in the diagram.
b) The initial position,P_0 of the pointer was read from the metre rule and recorded.
c) A standard mass, $M = 100\ g$ was suspended using the thread from the lower end of the spring.
d) The new position, P_1 of the pointer was read from the metre rule and recorded.

e) The extension, e, was determined from; $e = P_1 - P_0$
f) The steps (c) to (e) were repeated for different values of $M = 200, 300, 400, 500 \; and \; 600 \; g$
g) The results were tabulated/entered in a table.
h) The graph of M against e was plotted.
i) The slope, S of the graph was determined, which was the force constant of the spring.

POSSIBLE SOURCES OF ERRORS AND PRECAUTIONS

Error(s)	**Precaution(s)**
Parallax error in reading the marks from metre rule	Viewing the marks on the metre rule perpendicularly or at right angle.
Draught current (wind) from outside	Closing the windows and doors.

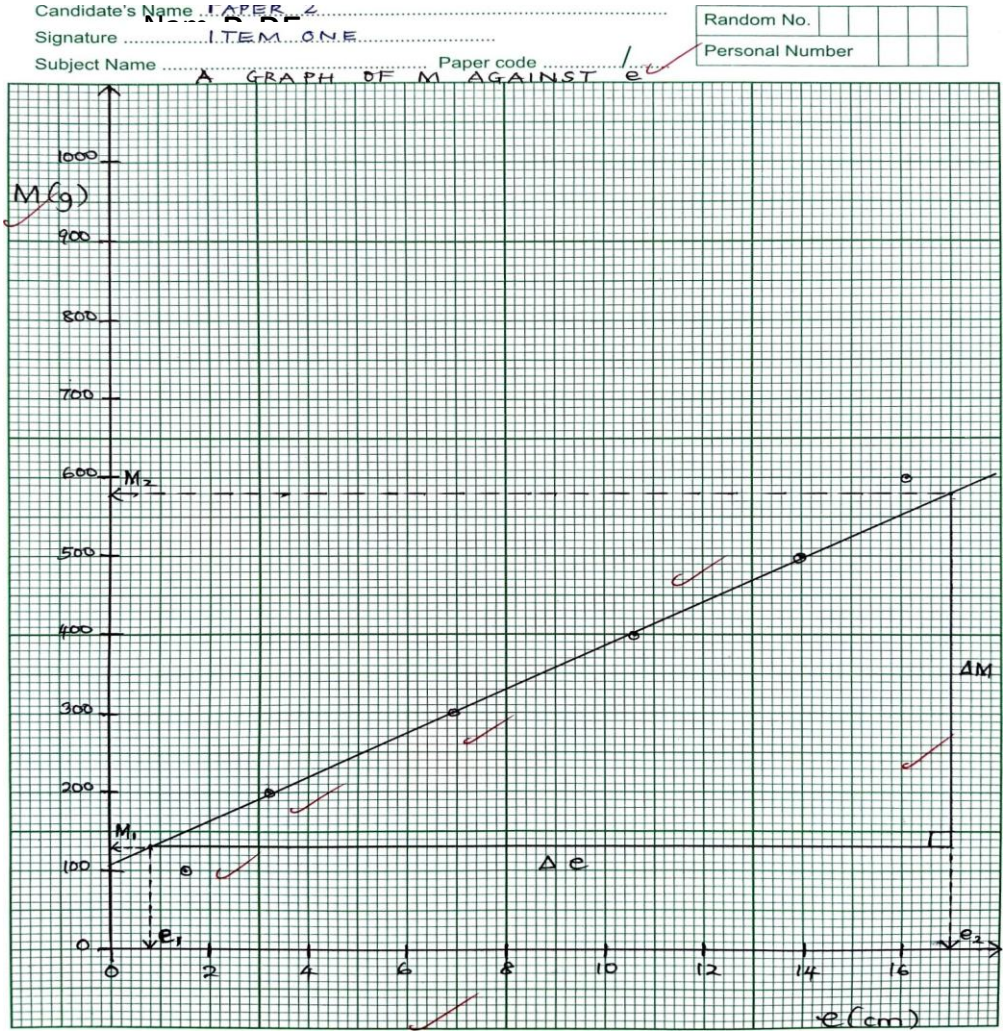
PRESENTATION OF DATA

$P_0 = 10.0cm$

Table of results

M(g)	**P_1(cm)**	**e(cm)**
100	11.5	1.5
200	13.2	3.2
300	16.9	6.9
400	20.6	10.6
500	24.0	14.0
600	26.1	16.1

DATA ANALYSIS



INTERPRETATION

From the graph;

$$slope, S = \frac{change\ in\ \boldsymbol{M}}{change\ \boldsymbol{e}}$$

$$S = \frac{580 - 130}{17.0 - 0.8}$$

$$S = \frac{450}{16.2}$$

$$S = 27.8\ Nm^{-1}$$

The force constant is twenty times; $K = 27.8\ x\ 20$

$$K = 556\ Nm^{-1}$$

Therefore, the force constant of the spring is $556\ Nm^{-1}$

CONCLUSION AND ADVISE

The force constant of $556\ Nm^{-1}$ is in the range of $200Nm^{-1}\ to\ 600Nm^{-1}$, so the hypothesis is valid/correct/accepted.

The mechanic/secretary can use the spring to replace the damaged one in the punching machine.

(if the force constant is between $200Nm^{-1}\ and\ 600Nm^{-1}$)

OR;

The force constant of $199\ Nm^{-1}\ or\ 601\ Nm^{-1}$, (any value outside the range), is not in the range of $200Nm^{-1}\ to\ 600Nm^{-1}$, so the hypothesis is invalid/wrong/rejected.

The mechanic/secretary should not use the spring to replace the damaged one in the punching machine.

(if the force constant is not between $200Nm^{-1}\ to\ 600Nm^{-1}$).

ITEM 2

AIM

An experiment to determine refractive index of glass and know if its critical angle is between $39\ ^0$and 42^0, so that it can be used for fixing in the window shutter.

VARIABLES

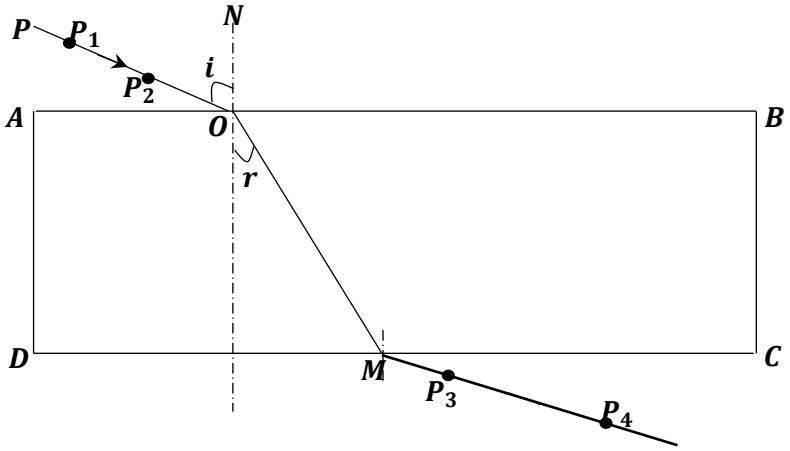
Variable	**Example (s)**
Independent	Angle of incidence, i
Dependent	Angle of refraction, r
Controlled	Light intensity, Amount of light in the room, Dimensions of the glass block.

HYPOTHESIS

The critical angle of the glass is between $39\ ^0$and 42^0.

PROCEDURE

i. Experimental setup/diagram;



ii. Steps in performing the experiment;

a) A plain sheet of paper was fixed on a soft board using drawing pins.

b) The glass block was placed in the centre of the paper.

c) The outline of the glass block was traced and labelled ABCD.

d) The glass block was removed, and a normal ON was drawn at point O, 2.0 cm from A.

e) A line PO was drawn at an angle $i = 10^0$ to the normal ON.

f) The glass block was replaced onto its outline.

g) Two optical pins, P_1 and P_2, were fixed on line PO.

h) On looking through side DC of the glass block, two more optical pins, P_3 and P_4, were fixed such that they appear in a line with the images of pins, P_1 and P_2. The glass block and pins P_3 and P_4 were removed.

i) A line RM was drawn through pin points P_3 and P_4 to cut DC at M.

j) The angle r was measured using the protractor and recorded.

k) The steps (e) to (k) were repeated for angles $i = 20, 30, 40, 50 \; and \; 60^0$.

l) The results were entered in a table, including values of $sin\, i$ and $sin\, r$.

m) A graph of $sin\, i$ against $sin\, r$ was plotted.

n) The slope n of the graph was determined, which was the refractive index of the glass block.

o) The critical angle, C was obtained from; $Sin\, C = \frac{1}{n}$

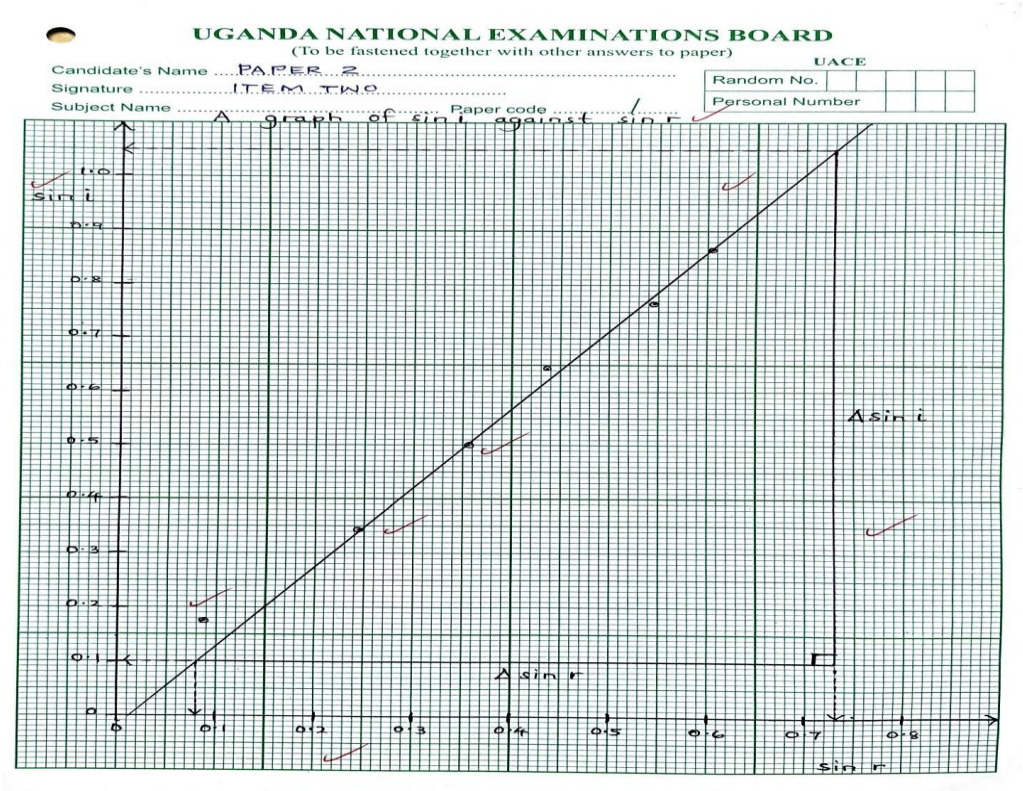
ERRORS AND PRECAUTIONS

Error (s)	Precaution(s)
Parallax error in reading the angles from the protractor	Viewing the angles perpendicularly or at right angle.
Too much (excess) light from outside	Closing the windows and doors/putting curtains.

DATA PRESENTATION

$i(°)$	r (°)	$Sin\ i$	$Sin\ r$
10	5	0.174	0.087
20	14	0.342	0.242
30	21	0.500	0.358
40	26	0.643	0.438
50	33	0.766	0.545
60	37	0.866	0.602

DATA ANALYSIS



DATA INTERPRETATION.

$$\boldsymbol{Slope\ , S = \frac{sin\ i}{sin\ r}}$$

$$S = \frac{1.05 - 0.10}{0.73 - 0.08}$$

$$S = \frac{0.95}{0.65}$$

$$S = 1.5$$

Critical angle, C is obtained as; $sin\ C\ = \frac{1}{n}$

$$= \frac{1}{1.5}$$

$$= \ 0.667$$

$$C\ = \ sin^{-1}\ (0.667)$$

$$= \ 41.80$$

$$= \ 42^0$$

Therefore, the critical angle is 42^0

CONCLUSION AND ADVISE

The critical angle of 42^0 is between $39\ ^0$and 42^0 so the hypothesis is valid/correct/accepted.

OR;

The critical angle of $39\ ^0$and 42^0 (any value outside the range), is not between $39\ ^0$and 42^0, so the hypothesis is invalid/wrong/rejected.

The builder can use the glass for fixing window shutter. (if C is between $39\ ^0$and 42^0)

OR;

The builder cannot use the glass for fixing window shutter. (if C is not between 39^0 and 42^0)

ITEM 3

Aim

A scientific investigation to determine if the internal resistance of the dry cell is in the range of 0.4 Ω and 1.0 Ω, so as to discover the quality of the dry cell or (the truth about the dry cell).

Variables

Variable	Example (s)

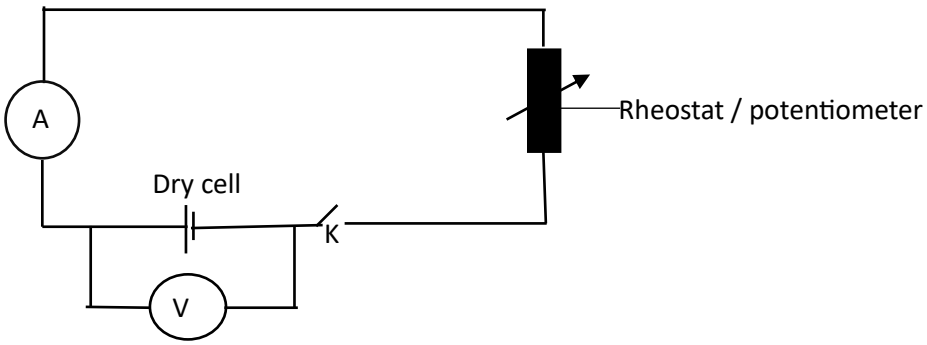
Independent	Ammeter reading/current, I. or Length/distance, x of the bare wire.
Dependent	Voltmeter reading/voltage, V / Ammeter reading/current, I
Controlled	Temperature of the wire, Contact resistance.

Hypothesis:

The internal resistance of the dry cell is not in the range of 0.4 Ω and 1.0 Ω,

Procedure

i) <u>Experimental setup/diagram;</u>



ii) **<u>Steps in performing the experiment;</u>**

The circuit was connected as shown in the diagram

The length/distance, d was measured from the metre rule such that d=10.0cm, and the switch was closed.

The ammeter reading, I and voltmeter reading, V were read from the ammeter and voltmeter respectively and recorded.

The steps (b) and (c) were repeated for different values of $d =$ $20.0, 30.0\ 40.0, 50.0\ and\ 60.0cm$

A graph of V against I was plotted.

The slope, S of the graph was determined.

The internal resistance, r of the cell was obtained as; $r = -S$

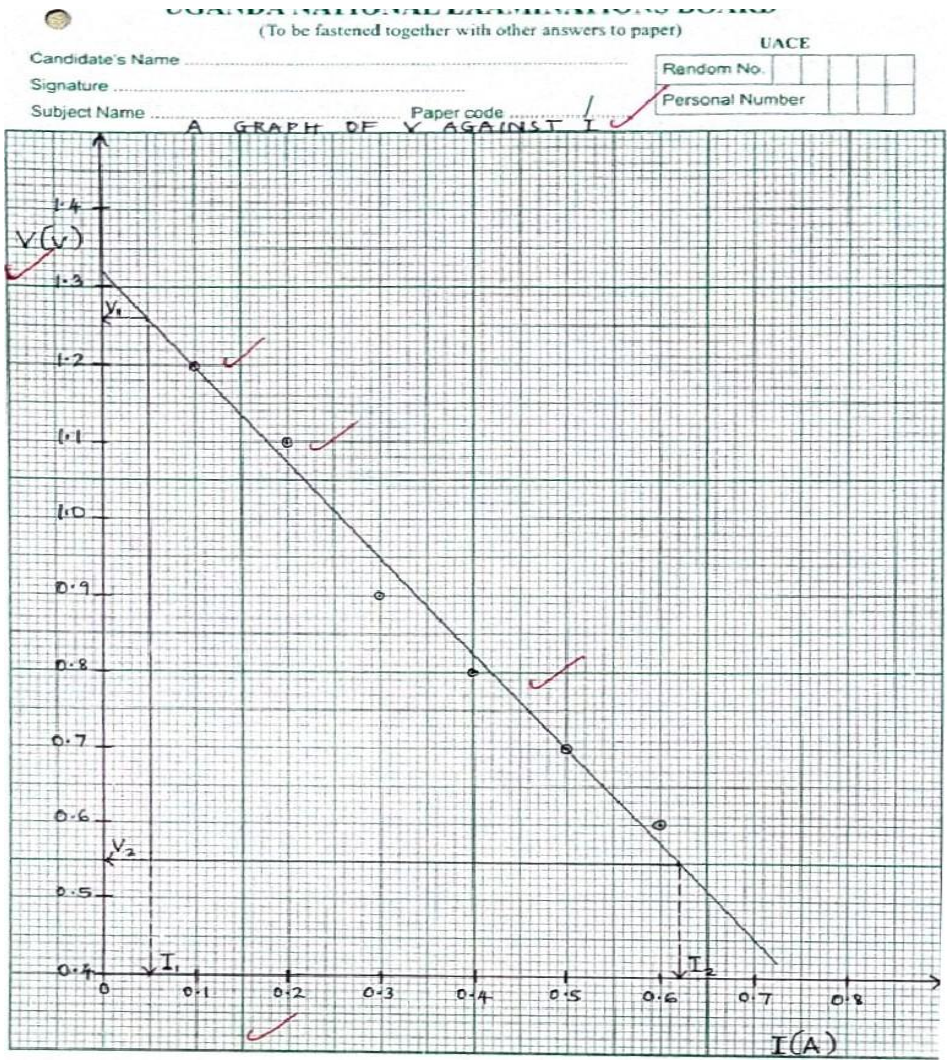
Possible sources of errors and precautions

S/n	Error	Precaution
1	Parallax error in reading the scales from the ammeter and voltmeter	Viewing the scales of ammeter and voltmeter perpendicularly or at right angle.
2	Loose connection of the wires	Tighten the wires

PRESENTATION OF DATA

d(cm)	I(A)	V(V)
60.0	0.10	1.20
50.0	0.20	1.10
40.0	0.30	0.90
30.0	0.40	0.80
20.0	0.50	0.70
10.0	0.60	0.60

DATA ANALYSIS



DATA INTERPRETATION

$$slope, S = \frac{change\ in\ V}{change\ in\ I}$$

$$S = \frac{0.55 - 1.26}{0.62 - 0.05}$$

$$S = \frac{-0.71}{0.57}$$

$$S = -1.2\ \Omega$$

The internal resistance,

$$r = -S$$
$$= -(-1.2)$$
$$= 1.2\ \Omega$$

Therefore, the internal resistance is 1.2 Ω

Conclusion and advise

The internal resistance of 1.2 Ω is not the range of 0.4 Ω and 1.0 Ω, so the hypothesis is valid/correct/accepted.

OR;

The internal resistance of 0.4 Ω or 1.0 Ω is the range of 0.4 Ω and 1.0 Ω,

(any value within the range), so the hypothesis is invalid/wrong/rejected.

It is true that the dry cells have higher internal resistance/ the dry cells are of poor quality. (if the internal resistance is more than 1.0 Ω)

OR;

It is not true, instead the dry cells have the standard internal resistance/ the dry cells are of good quality. (if the internal resistance is between 0.4 Ω and 1.0 Ω)